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Studierichting: Informatica
Oefeningenreeks 5F nr. 2.5

$$r_1 = 1 \text{ en } r_2^2 = \cos 2\theta$$

opp. r_1 :

$$\frac{1}{2} \int_0^\pi 1 d\theta = \pi$$

opp. r_2 :

$\cos 2\theta$ bestaat tussen $-\frac{\pi}{4}$ en $\frac{\pi}{4}$ en gespiegeld tussen $-\frac{3\pi}{4}$ en $\frac{3\pi}{4}$

Dit kan men vereenvoudigen tot:

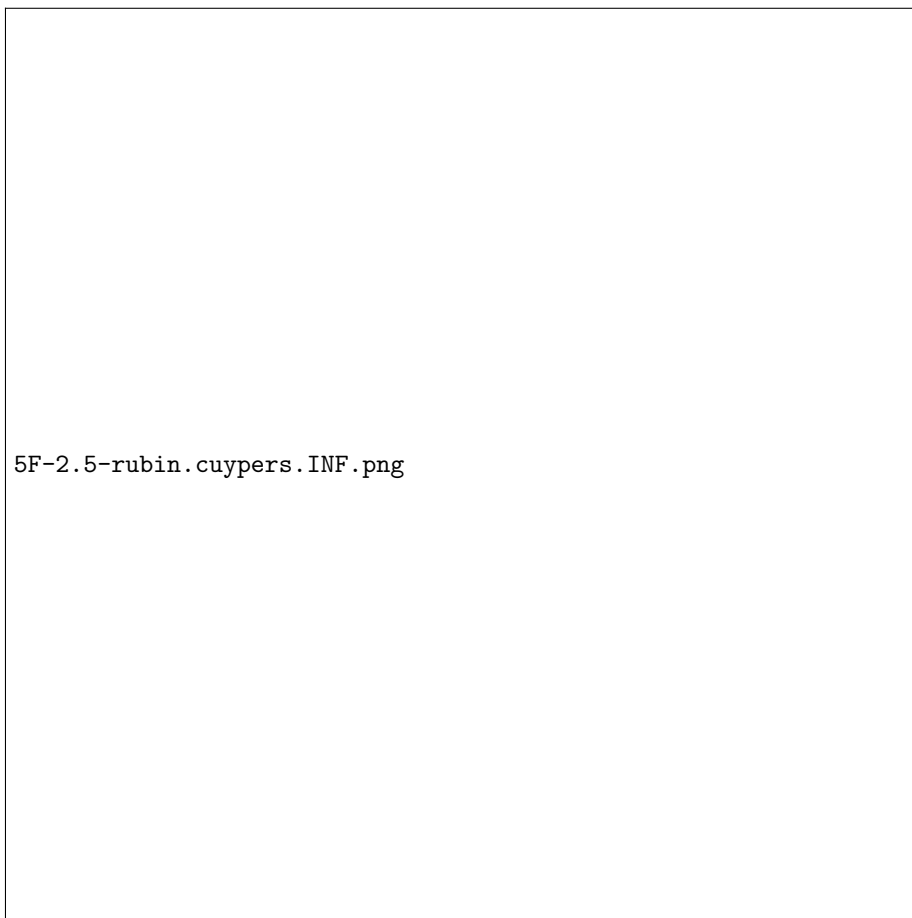
$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta d\theta$$

$$= \left[\frac{\sin 2\theta}{2} \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$$

$$= 1$$

$$\Rightarrow \text{opp. } (r_1 - r_2) = \pi - 1$$

References



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